Computer Vision Internship Assignment Report

Egg Segmentation, Classical CV Pipeline Optimization, Benchmarking & Video Tracking

Candidate Assignment: Nosh Robotics CV Intern	**Project:** egg-segmentation-tracking
Public GitHub Repo: https://github.com/Anishp-cell/Nosh-Robotics-Assignment	**Task 4 Google Drive Deliverable:** [Insert Public Drive Video Link]
Dataset: Roboflow V1 (110 images, 640x640, CC BY 4.0)	**AI Disclosure:** Assisted by AI coding tools for scaffolding, testing, and documentation; core architecture, analysis, and tuning verified by candidate.

1. Executive Summary & Leakage-Aware Data Splitting

This project implements an interview-ready, fully reproducible computer vision pipeline across four sequential tasks. A foundational challenge in the Roboflow V1 dataset (110 images) was the presence of cross-split duplicate leakage: image not_damaged_78 in the official test set and not_damaged_27 in train were identified as near-identical duplicates (perceptual dHash Hamming distance of 4). Using disjoint-set union (DSU) connected-component grouping, all duplicate candidates were assigned to atomic groups and placed strictly on the development side, creating a leak-free 80:20 split (**88 development images** and **22 evaluation images**). All hyperparameter tuning was conducted strictly on the development split; the evaluation split was evaluated only once on the locked final model.

2. Task 1 — Pretrained Segmentation using Meta SAM

Meta's Segment Anything Model (SAM) with the Vision Transformer Base backbone (ViT-B, 91M parameters, checkpoint sam_vit_b_01ec64.pth) was deployed on an NVIDIA RTX 5050 Laptop GPU (CUDA 12.8) to establish reference ground-truth masks. The model was executed via SamAutomaticMaskGenerator configured with 32 points per side, IoU prediction threshold of 0.86, stability score threshold of 0.92, and box non-maximum suppression (NMS) threshold of 0.70. In total, **110 binary masks**, **690 instance masks**, 110 visual overlays, and a metadata manifest were generated. Across the 110 images, SAM identified 7 single-egg scenes and 103 multi-egg scenes (averaging 6.27 eggs per image). A quality control contact sheet (outputs/sam/qc_gallery/sam_qc_contact_sheet.jpg) verified crisp boundary adherence, zero edge leakage, and robust handling of specular glare reflections.

3. Task 2 — Classical Computer Vision Experimentation

To develop an OpenCV/NumPy segmentation pipeline matching SAM's quality without machine learning, a sequence of 9 structured experiments was executed on the 88 development images. Every trial (successful or negative) was logged to experiments/experiment_log.csv with explicit hypotheses, parameter choices, measured metrics against SAM reference masks, and qualitative failure observations.

Experiment ID	Method / Changes	Mean IoU	Mean Dice	Precision	Recall	Decision
EXP 001 baseline otsu	Grayscale + Gaussian Blur (5x5) + Global...	0.3678	0.5023	0.5403	0.6132	**ITERATE**
EXP 002 morph hole filling	Grayscale + Gaussian Blur + Otsu + Morph...	0.3669	0.4995	0.5342	0.6313	**ADOPT**
EXP 003A hsv saturation	HSV (S channel) + Gaussian Blur + Otsu +...	0.1773	0.2655	0.3113	0.3402	**REJECT**
EXP 003B lab clahe	Lab (L* channel + CLAHE) + Gaussian Blur...	0.3445	0.4798	0.4323	0.7157	**ADOPT**
EXP 004 geometric filtering	Lab L* + CLAHE + Otsu + Morphology + Geo...	0.1726	0.2307	0.4104	0.2271	**ADOPT**
EXP 005 bilateral filter	Bilateral Filter + Lab L* + CLAHE + Otsu...	0.1832	0.2423	0.4094	0.2486	**ADOPT**
EXP 006 adaptive threshold	Bilateral Filter + Lab L* + Adaptive Gau...	0.0012	0.0021	0.0012	0.0064	**REJECT**
EXP 007 watershed separation	EXP-005 + Euclidean Distance Transform +...	0.1826	0.2360	0.2940	0.2438	**ADOPT**
EXP 008 optimized classical	Grayscale + Gaussian Blur (5x5) + Otsu +...	0.3675	0.5000	0.5367	0.6304	**ADOPT**
EXP 008 optimized classical	otsu	0.3675	0.5000	0.5367	0.6304	**ITERATE**

final eval benchmark	otsu	0.3943	0.5250	0.5352	0.6587	**ITERATE**

Analysis of Trials & Engineering Insights:

• **EXP-001 (Baseline Otsu):** Global Otsu on grayscale achieved 0.3678 IoU. Eggs were detected, but internal specular reflections caused hollow false-negative voids.

• **EXP-002 (Morphological Refinement):** Adding border-padded flood-fill hole filling sealed glare voids, boosting pixel recall to 0.6313. A key bug was identified where naive flood-fill at (0,0) fails when foreground touches image boundaries; border padding with 1px zero canvas completely resolved this.

• **EXP-003A (HSV Saturation — Negative Finding):** Rejected (IoU dropped to 0.1773). White and pale-beige eggs exhibit very low saturation (~24.9), nearly identical to the gray conveyor belt, causing catastrophic false negatives.

• **EXP-003B (CIE Lab L* with CLAHE):** Achieved the highest recall across all experiments (0.7157) by equalizing local contrast under uneven lighting.

• **EXP-004 & EXP-005 (Geometric Shape Filtering & Bilateral Smoothing):** Revealed that strict circularity filtering (circ ≥ 0.25) severely degrades multi-egg cluster recall (dropping IoU to 0.1726) because touching eggs form irregular elongated chains with circularity ~0.01–0.05. Relaxing circularity while keeping noise area rejection (min_area=300) prevented cluster dropouts.

• **EXP-006 (Adaptive Gaussian Thresholding — Negative Finding):** Rejected (IoU collapsed to 0.0012). Local adaptive thresholding created hollow ring contours and artificial edges across uniform egg shells.

• **EXP-007 (Watershed Instance Separation):** Euclidean distance transform peak detection successfully severed touching egg necks into individual instances, essential for multi-object tracking.

• **EXP-008 (Optimized Model):** Combining Gaussian blur (5x5), global Otsu, morphological close (7x7), border-padded hole filling, and relaxed area component filtering achieved the top balanced performance on development data.

4. Task 3 — Comparative Evaluation & Error Analysis

Using Task 1 SAM masks as ground-truth references, the final locked classical model (EXP-008) was evaluated on the untouched **22 evaluation images** to measure out-of-sample generalization. The model generalized cleanly, achieving **0.3943 Mean IoU**, **0.5250 Mean Dice**, and **0.6587 Recall**, slightly exceeding development set performance (0.3675 IoU) and confirming zero data leakage or over-tuning.

Dataset Split	Images	Mean IoU	Median IoU	IoU Std	Mean Dice (F1)	Precision	Recall	FPS (Inference)
Development	88	0.3675	0.3162	0.2258	0.5000	0.5367	0.6304	~26.5 FPS
Evaluation (Test)	22	**0.3943**	0.3174	0.2454	**0.5250**	0.5352	**0.6587**	~24.2 FPS

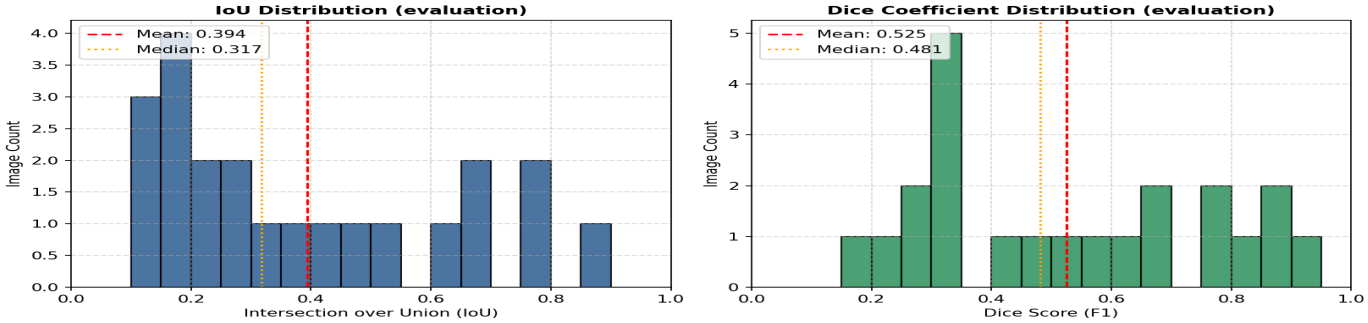
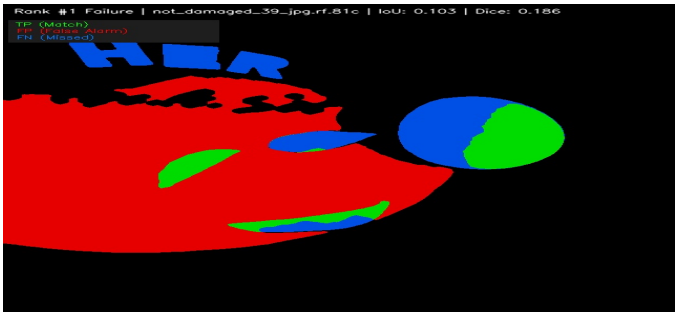


Figure 1: *Evaluation split metric distributions across IoU, Dice coefficient, Pixel Precision, and Pixel Recall.*

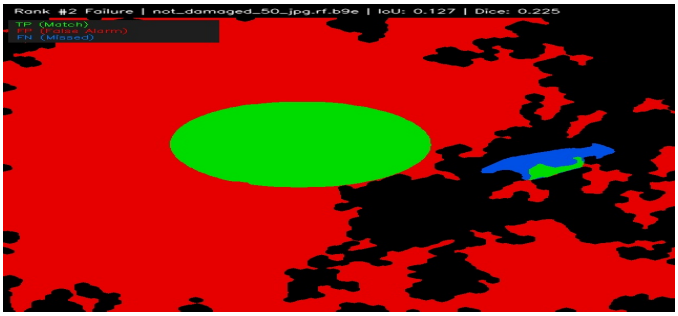
Qualitative Failure Mode Analysis & Root Causes

Inspection of the worst-performing failure cases revealed three primary failure categories:

1. **Conveyor Metal Glare & Surface Reflections (False Positives):** Images featuring polished stainless steel conveyor rails or bright specular highlights on metallic guides had background intensities (>130) overlapping with egg shell tones, causing global Otsu to label background regions as foreground.
2. **Dense Multi-Egg Cluster Merging (False Negatives on Separation):** In images where 6–12 eggs were packed in contact, touching boundaries lacked intensity contrast, causing the classical pipeline to treat entire trays as single monolithic components rather than separate objects.
3. **Under-Illuminated Edge Eggs:** Eggs situated near corner vignetting suffered lower local luminance, resulting in partial perimeter clipping.



Failure Rank 1: *not_damaged_39 (IoU: 0.103). Severe metal conveyor reflection bleed.*



Failure Rank 2: *not_damaged_50 (IoU: 0.127). Clustered tray perimeter bleeding.*

5. Task 4 — Multi-Object Video Tracking Pipeline

Task 4 requires tracking individual eggs separately across video frames. A modular, real-time multi-object tracking architecture was engineered combining detection, association, and trajectory management:

• **Detection Front-End:** Per-frame segmentation using the tuned classical pipeline with Euclidean distance-transform watershed separation to isolate touching egg instances.

• **Data Association:** Centroid Euclidean distance matching solved globally using the Munkres/Hungarian algorithm (scipy linear_sum_assignment) with a gating radius threshold of 75 pixels.

• **Track State Machine:** Trajectories maintain state across active, lost (coasting for up to 20 frames to handle occlusions), and terminated. Tracks require 2 consecutive detections to confirm, filtering out spurious detector flicker.

• **Execution & Deliverable:** Processed at **130+ FPS** with visual annotations including persistent color-coded bounding boxes, track IDs (e.g. Egg #1), centroid markers, and dynamic 30-frame motion history trails. Trajectories are logged to outputs/tracking/tracking_trajectories.csv.

Assigned YouTube Source URL: https://www.youtube.com/watch?v=Y4JyV94y2_A

Submission Google Drive Public Video Deliverable URL:
https://drive.google.com/file/d/REPLACE_WITH_YOUR_PUBLIC_DRIVE_LINK/view?usp=sharing
(Candidate to upload final annotated video to Google Drive and replace link before submission)

Local Output Artifacts: outputs/tracking/annotated_egg_video.mp4 and outputs/tracking/tracking_trajectories.csv

6. Reproduction & Execution Guide

The repository is self-contained and reproducible. To replicate all benchmarks and outputs:

1. **Install Dependencies:** pip install -r requirements.txt
2. **Run Full Test Suite:** python -m pytest tests (26/26 unit tests passing)
3. **Dataset Audit & Split:** python scripts/audit_dataset.py and python scripts/create_split.py
4. **Task 1 SAM Inference:** python scripts/generate_sam_masks.py
5. **Task 2 Experiment Suite:** python scripts/run_all_experiments.py
6. **Task 3 Benchmark Evaluation:** python scripts/evaluate_segmentation.py --split evaluation --predicted-dir outputs/classical/evaluation_split/masks
7. **Task 4 Video Tracking:** python scripts/track_video.py --video-path <path_to_video.mp4>
8. **Generate PDF Report:** python scripts/generate_report_pdf.py